

What was rederived, what was reproduced, and what was only called

A verification record for the finite-coupling extremality-shift calculation

28 September 2026

Abstract

This is not a physics paper; it is an audit trail. It states, claim by claim, which parts of the accompanying calculation were derived here, which were computed here and checked against something external, which were copied from a paper and verified against that paper, and which were simply used on the author's authority. Of 58 entries, 47 carry a machine-checked verdict and 11 are listed as called and not reproduced. 0 checks fail. The external anchors are the Myers–Perry solution in the vacuum limit, the Boulware–Deser solution at finite coupling and zero spin, the perturbative potential of Wu and Lü at zero coupling, the extremal near-horizon branch of Brihaye, Kleihaus, Kunz and Radu at finite coupling, and the extremality shift of Ma, Li and Lü. Everything is regenerated by `work/verification_suite.py`, which is what produced the tables below.

How to read this

Four kinds of entry appear, and the distinction between them is the reason the document exists. A calculation that agrees with itself has shown nothing; the question is always what it was tested against, and with what.

Rederived analytically. Redone here symbolically, with the difference asserted to be exactly zero. These need no tolerance because nothing numerical enters: either the expressions are identical or they are not.

Reproduced numerically. Produced by the spectral solver and compared with a closed form or a published equation. Each carries the tolerance its assertion used, and figures 1–4 draw the deviation against that tolerance so the margin is visible rather than asserted.

Transcription checked against the source. Every convention this work imports — a coupling normalisation, a charge normalisation, a factor in a near-horizon metric — read back from the source PDF. These were previously the weakest link: the manuscript's own referee record listed the literature attributions as unverified because no source PDFs were present. They now are, under `papers/`, and the checks below read them.

Called, not reproduced. Used and not re-derived. Listing these is the only way the three categories above mean anything.

The three conversions that everything else depends on

Three numbers relate this work’s conventions to the literature, and a mistake in any of them would move every finite-coupling comparison without breaking any internal check. Each is now verified twice, by independent routes.

The coupling. The source writes its action as $I = (16\pi G)^{-1} \int \sqrt{-g} (R + \frac{\alpha_{\text{ref.}}}{4} \mathcal{L}_{\text{GB}})$ while this work writes $R + \alpha \mathcal{L}_{\text{GB}}$, so $\alpha_{\text{ref.}} = 4\alpha$. That is read directly from eq. (2.1) of arXiv:1010.0860. Independently, the ratio of the Gauss–Bonnet piece of the published entropy (3.10) to its Einstein piece equals the ratio our Jacobson–Myers form gives *only* if $\alpha_{\text{ref.}} = 4\alpha$; that is checked symbolically. Two routes, one answer.

The angular momentum. The source’s J is the momentum in *each* plane, not the total, which is the sentence introducing its eq. (3.6). Its normalisations $E = -3V_3 U/16\pi G$ and $J = V_3 W/8\pi G$ with $V_3 = 2\pi^2$ reduce to the $-3\pi U/8$ and $\pi W/4$ used here.

The near-horizon k . The source’s near-horizon metric carries $\sigma_3 + 2kr dt$ where this work writes $\sigma_3 + kr dt$, so $k = 2k_{\text{ref.}}$ and, since $J = \partial E/\partial k$, $J_{\text{ref.}} = 2J$. This is what reconciles the source’s $S_{\text{ext}} = \pi J_{\text{ref.}}$ at zero coupling with the $S_{\text{ext}} = 2\pi J$ obtained here, and both with extremal Myers–Perry, $S = 2\pi^2 a^3$ and $J = \pi a^3$. Without the factor this would look like a discrepancy of two; with it, the three statements are one statement.

The ledger

Rederived analytically

Redone here with sympy and the difference asserted to vanish identically. No tolerance and no solver output enters.

claim	source	result	tolerance	verdict
Psi_static = (3pi/4)(4a-3)/(1+4a) from dM/da - T dS/da	Boulware-Deser charges; manuscript eq. (15)	–	–	pass
Psi_static -> -9pi/4 as alpha -> 0	manuscript eq. (15)	–	–	pass
Psi_static changes sign at alpha/r_H^2 = 3/4	manuscript sec. 4.1	–	–	pass
Smarr 2M = 3TS + 2*alpha*Psi closes on Boulware-Deser at J=0	manuscript eq. (8) on eq. (14)	–	–	pass
static t_H = sqrt(1+2a)/(1+4a) follows from eq. (10)	manuscript sec. 2.3	–	–	pass
Euler on the listed weights gives 2M = 3TS + 3 sum(Omega_i J_i) + 2 alpha Psi	manuscript eq. (8)	–	–	pass
its equal-spin case is the printed 2M = 3TS + 6 Omega_H J + 2 alpha Psi	manuscript eq. (8)	–	–	pass

claim	source	result	tolerance	verdict
differentiated Smarr minus twice the first law is the printed Gibbs-Duhem constraint	manuscript sec. 3.1	–	–	pass
$q = r_- \Omega_H = a/r_-$ from $\Omega_H = a/(r_+^2 + a^2)$, $r_-^2 = r_+^2 + a^2$	manuscript eq. (9)	–	–	pass
$u = q^2/(1-q^2)$ equals a^2/r_+^2	manuscript eq. (9)	–	–	pass
equal-spin Myers-Perry is extremal at $r_+ = a$ ($d\mu/d r_+ = 0$)	Myers-Perry horizon function	–	–	pass
hence extremality sits at $q = 1/\sqrt{2}$	manuscript sec. 2.1	–	–	pass
$M = 3\pi r_-^2/[8(1-q^2)]$ is the same as $3\pi \mu_{MP}/8$	manuscript appendix A	–	–	pass
$\Psi_0(u=0) = -9\pi/4$, matching the $\alpha \rightarrow 0$ limit of Ψ_{static}	manuscript eq. (17) against eq. (15)	–	–	pass
$\Psi_0(u=1) = \pi$, matching the published linear-order slope	manuscript eq. (17) against arXiv:2009.00015	–	–	pass
Ψ_0 sign zero at $q = 7-2\sqrt{10}$ mapped through eq. (9)	manuscript sec. 4.2	6.35×10^{-1}	1.00×10^{-9}	pass
our metric ansatz is identically eq. (2.8) of 1010.0860 with $g = r^2$	manuscript eq. (4) against arXiv:1010.0860 eq. (2.8)	–	–	pass
the published entropy (3.10) equals our Jacobson-Myers form iff $\alpha_{paper} = 4\alpha$	manuscript eq. (7) against arXiv:1010.0860 eq. (3.10)	–	–	pass
$dS/d\alpha$ at fixed (M, J) equals $-\Psi/T$, from the extended first law	manuscript sec. 3.2	–	–	pass

Reproduced numerically

Computed by this project's solver and compared against a closed form or a published equation. Each carries the tolerance its assertion used.

claim	source	result	tolerance	verdict
the $\alpha=0$ mass reproduces the Myers-Perry closed form	Myers-Perry, $M = 3\pi r_-^2/[8(1-q^2)]$	3.27×10^{-13}	1.00×10^{-10}	pass
the $\alpha=0$ horizon squashing reproduces the Myers-Perry $1/(1-q^2)$	Myers-Perry	8.30×10^{-14}	1.00×10^{-8}	pass
the static mass reproduces the Boulware-Deser closed form at finite coupling	Boulware-Deser	3.20×10^{-11}	1.00×10^{-8}	pass
the static temperature reproduces the Boulware-Deser closed form at finite coupling	Boulware-Deser	1.08×10^{-13}	1.00×10^{-8}	pass

claim	source	result	tolerance	verdict
the static entropy reproduces the Boulware-Deser closed form at finite coupling	Boulware-Deser	1.00×10^{-16}	1.00×10^{-8}	pass
the static potential reproduces the Boulware-Deser closed form at finite coupling	Boulware-Deser	2.60×10^{-9}	1.00×10^{-8}	pass
the $\alpha=0$ response reproduces the published perturbative potential	arXiv:2405.04576 eq. (33), mapped by manuscript eq. (17)	2.86×10^{-8}	1.00×10^{-6}	pass
our near-horizon solution satisfies published eq. (4.11) as printed	arXiv:1010.0860 eq. (4.11)	9.17×10^{-14}	1.00×10^{-10}	pass
our near-horizon solution satisfies published eq. (4.12) as printed	arXiv:1010.0860 eq. (4.12)	6.66×10^{-16}	1.00×10^{-10}	pass
the near-horizon branch gives $S_{\text{ext}} = 2 \pi J$ at $\alpha=0$, the extremal Myers-Perry value	Myers-Perry; arXiv:1010.0860 eq. (4.13) after $k_{\text{paper}} = k/2$	1.11×10^{-12}	1.00×10^{-10}	pass
the extrapolated bulk entropy reproduces the near-horizon curve at finite coupling	arXiv:1010.0860 sec. 4.2	4.06×10^{-6}	1.00×10^{-4}	pass
the extrapolated $\mu(0)$ reproduces the published $(3/2) \pi^{1/3}$	arXiv:2009.00015	4.56×10^{-11}	1.00×10^{-8}	pass
the extrapolated $\Psi_{\text{ext}}(0)$ reproduces the published π	arXiv:2009.00015	1.19×10^{-7}	1.00×10^{-5}	pass
the vacuum extremal state returns the normalisation value $j = 1$	arXiv:2303.12471 scaled spin	3.11×10^{-11}	1.00×10^{-6}	pass
the Smarr relation closes on every accepted rotating solution	manuscript eq. (8)	1.64×10^{-7}	1.00×10^{-6}	pass
the first law closes in the spin direction at fixed coupling	manuscript eq. (2)	1.35×10^{-8}	1.00×10^{-6}	pass
every accepted state satisfies $G + \alpha H = 0$ off the solver grid	Einstein-Gauss-Bonnet field equations	9.76×10^{-9}	1.00×10^{-8}	pass
the extremal intercept survives two coarser radial resolutions at every coupling	this work, resolution study	1.50×10^{-3}	1.00×10^0	pass

Transcription checked against the source

A convention, normalisation or formula this work takes from a paper, checked against the text of that paper as downloaded in `papers/`.

claim	source	result	tolerance	verdict
the source action carries $\alpha/4$ on L_{GB} , so $\alpha_{\text{paper}} = 4 \alpha$	arXiv:1010.0860 eq. (2.1)	—	—	pass
the source charges are $E = -3 V_3 U/16 \pi G$ and $J = V_3 W/8 \pi G$ with $V_3 = 2 \pi^2$	arXiv:1010.0860 eq. (3.6)	—	—	pass
the source J is the momentum in each plane, not the total	arXiv:1010.0860 sec. 3.2	—	—	pass

claim	source	result	tolerance	verdict
the source near-horizon metric carries $2k$, so our k is twice theirs	arXiv:1010.0860 eq. (4.6)	–	–	pass
the source entropy has an Einstein piece and a GB piece of the printed form	arXiv:1010.0860 eq. (3.10)	–	–	pass
the source quotes an outer ergosurface radius at the coupling we compare with	arXiv:1010.0860 sec. 4.1	–	–	pass
the published extremal mass is $M = (3/2) \pi^{1/3} J^{2/3} + \pi \alpha + O(\alpha^2)$	arXiv:2009.00015	–	–	pass
the source uses the same $1/16 \pi$ action normalisation, with $G = 1$	arXiv:2405.04576 eq. (25)	–	–	pass
their $\Delta M = (D-4) \Delta G$ at $D=5$ with equal spins is exactly our eq. (17)	arXiv:2405.04576 eq. (33)	–	–	pass
the source treats a general quadratic-curvature correction on a Ricci-flat background	arXiv:2405.04576	–	–	pass

Called, not reproduced

Used on the authority of the source. Listed so that it cannot be mistaken for anything above.

what	source	status
The Myers-Perry solution itself	Myers & Perry, Annals Phys. 172 (1986) 304	Its closed forms are used as an oracle and every one of them is checked against our solver, but the solution is not rederived here. Not on arXiv, so no PDF is in papers/.
The Boulware-Deser solution itself	Boulware & Deser, PRL 55 (1985) 2656	Same status: the closed forms for M , T and S are the oracle for the static limit, and we differentiate them to get Ψ , but the solution is not rederived. Not on arXiv.
Lovelock’s theorem	Lovelock, J. Math. Phys. 12 (1971) 498	Cited for the field equations being second order. Not reproduced. Not on arXiv.
The Wald entropy formula	arXiv:gr-qc/9307038, arXiv:gr-qc/9403028	The Noether-charge form is cited, not derived. Its value on this horizon is checked against the independent derivation in arXiv:1010.0860.
The Jacobson-Myers form for Lovelock horizons	arXiv:hep-th/9305016	Cited as the form Wald entropy takes here; the resulting entropy is then verified numerically against the published near-horizon branch.

what	source	status
Sen's entropy function formalism	arXiv:0708.1270	The formalism is cited. Our use of it is rederived symbolically and its solution checked against the full field equations component by component, but the formalism itself is not re-justified.
The domain bounds $x < 1$ and $j \leq 1$	arXiv:2303.12471	Quoted and compared against, not rederived. Our samples lie inside them.
The Goon-Penco universality argument	arXiv:1909.05254	Only the first-law identity is used, and that is rederived here. The universality claim itself is cited and not tested.
Causality constraints on higher-curvature couplings	arXiv:1407.5597	Cited in the discussion as a reason to separate the classical truncation from an ultraviolet completion. Nothing is computed from it.
The extended first law for Lovelock couplings	arXiv:1005.5053, arXiv:2404.16981	Cited as the source of the Ψ α term. The term is then used and its consequences checked internally, but the covariant-phase-space derivation is not repeated.
The outer ergosurface radius 1.104 at $\alpha = 1/2$	arXiv:1010.0860	Compared against and does NOT reproduce: our reconstruction gives 1.102101. Unresolved; recorded in the manuscript discussion as an open point.

Sources held locally

Every paper the calculation leans on that exists on arXiv is in `papers/`, and the transcription checks above read them from there. The three pre-arXiv sources — Myers and Perry (1986), Boulware and Deser (1985) and Lovelock (1971) — are not available that way and are listed as called.

file	size
0708.1270-sen-entropy-function-attractors.pdf	1835 kB
1005.5053-kastor-ray-traschen-smarr-lovelock.pdf	262 kB
1010.0860-brihaye-kleihaus-kunz-raduequal-spin-egb-d5.pdf	396 kB
1306.2517-kunduri-lucietti-near-horizon-classification.pdf	883 kB
1407.5597-camanho-edelstein-maldacena-zhiboedov-causality.pdf	706 kB
1901.11535-reall-santos-higher-derivative-kerr-thermodynamics.pdf	474 kB
1909.05254-goon-penco-entropy-extremality-relation.pdf	171 kB
2009.00015-ma-li-lu-d5-egb-mass-angular-momentum-extremality.pdf	281 kB
2303.12471-kleihaus-kunz-radu-phases-rotating-d5-egb.pdf	301 kB
2404.16981-neri-liberati-covariant-phase-space-lanczos-lovelock.pdf	877 kB
2405.04576-wu-lu-quadratic-curvature-rotating-thermodynamics.pdf	301 kB
2512.23797-ma-lu-quadratic-curvature-correction-5d-myers-perry.pdf	245 kB
gr-qc-9307038-wald-black-hole-entropy-noether-charge.pdf	142 kB
gr-qc-9403028-iyer-wald-noether-charge-dynamical-entropy.pdf	307 kB
hep-th-0601001-arkani-hamed-motl-nicolis-vafa-weak-gravity.pdf	213 kB
hep-th-0606100-kats-motl-padi-higher-order-extremal-mass-charge.pdf	174 kB
hep-th-9305016-jacobson-myers-entropy-higher-curvature.pdf	134 kB

What this record does not establish

Agreement with an external result bounds the error at the point of comparison and nowhere else. The vacuum and static checks are exact but sit at the edges of the family — zero coupling and zero spin — and the near-horizon check constrains the entropy, not the mass. No external result exists for the rotating mass at finite coupling, which is precisely the quantity the accompanying manuscript reports; that number rests on the solver, its convergence behaviour and the zero-temperature extrapolation, and on nothing published. The tolerances quoted here are the ones the assertions used, chosen to be loose enough to pass reliably and not as estimates of the accuracy achieved: the achieved deviations are typically several orders of magnitude smaller, which is what the lower panels of the figures show.

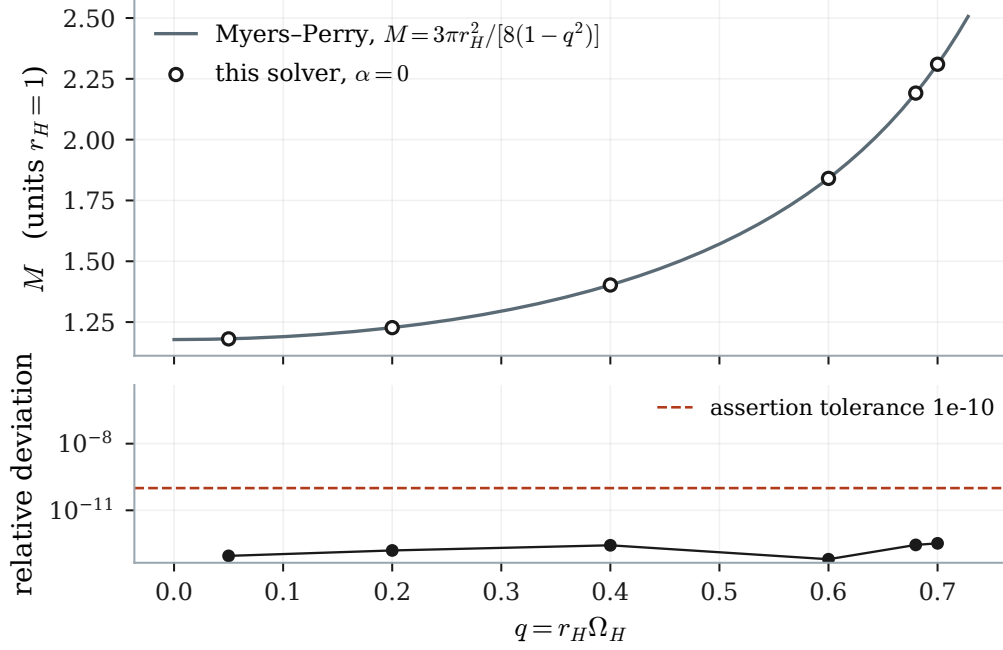


Figure 1. The vacuum limit against Myers–Perry. Upper: the mass of the solver’s $\alpha = 0$ solutions against the closed form $M = 3\pi r_H^2 / [8(1 - q^2)]$, which follows from $3\pi\mu/8$ and the coordinate map $r^2 = \rho^2 + a^2$ rederived in this report. Lower: the relative deviation, against the tolerance the assertion used. The comparison runs to $q = 0.7$, past the vacuum extremal spin $1/\sqrt{2}$ being approached.

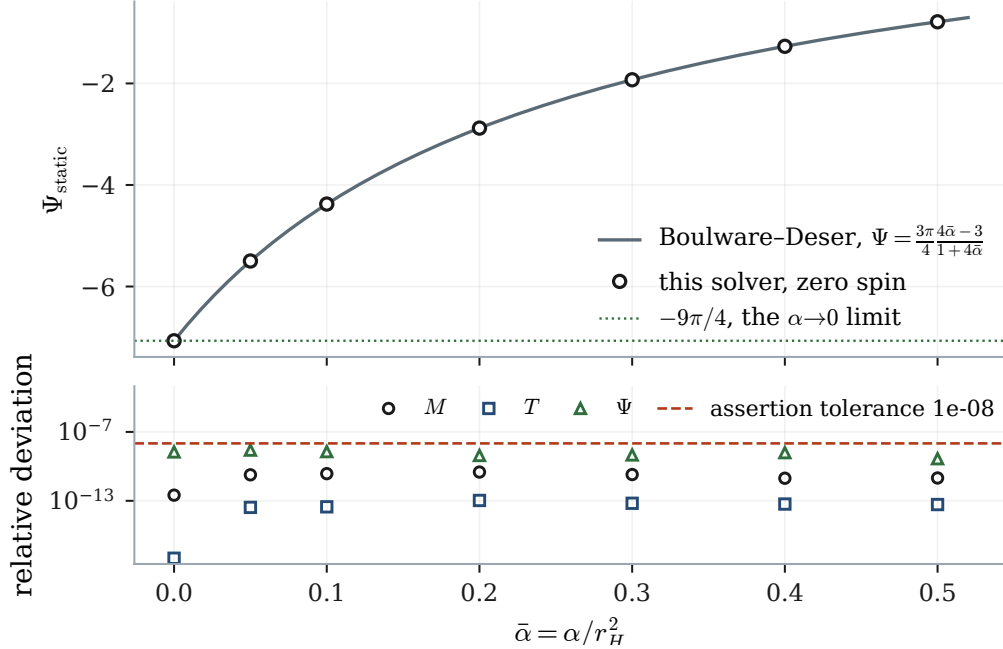


Figure 2. The static limit against Boulware–Deser, at finite coupling. Upper: the potential Ψ measured at zero spin against the closed form derived in this report by differentiating the Boulware–Deser charges. This is the one exact check available at $\alpha \neq 0$. The dotted line is the $\alpha \rightarrow 0$ value $-9\pi/4$, which the independent perturbative formula also returns. Lower: relative deviations in M , T and Ψ . The entropy is reproduced to machine precision and is omitted from the lower panel for that reason.

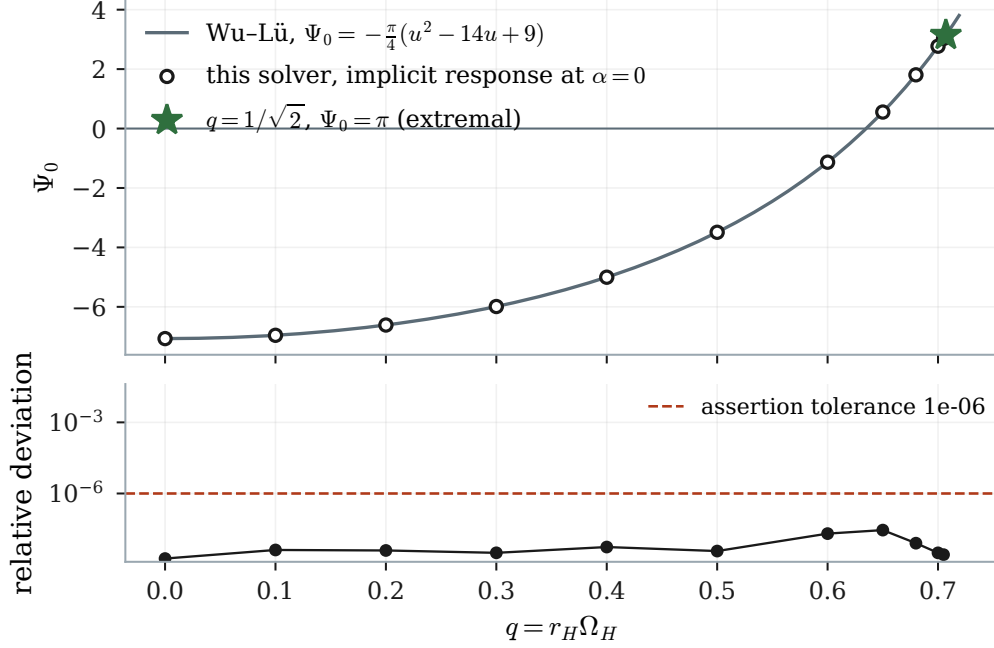


Figure 3. The zero-coupling response against the published perturbative potential. Upper: the implicit numerical response evaluated on the Myers–Perry background against $\Psi_0(q)$, which this report re-derives by specialising the general- D result of Wu and Lü to $D = 5$ with equal spins rather than by copying it. The star marks the extremal endpoint $\Psi_0 = \pi$, which is the published linear-order extremality shift and was not used in obtaining the curve. Lower: relative deviation.

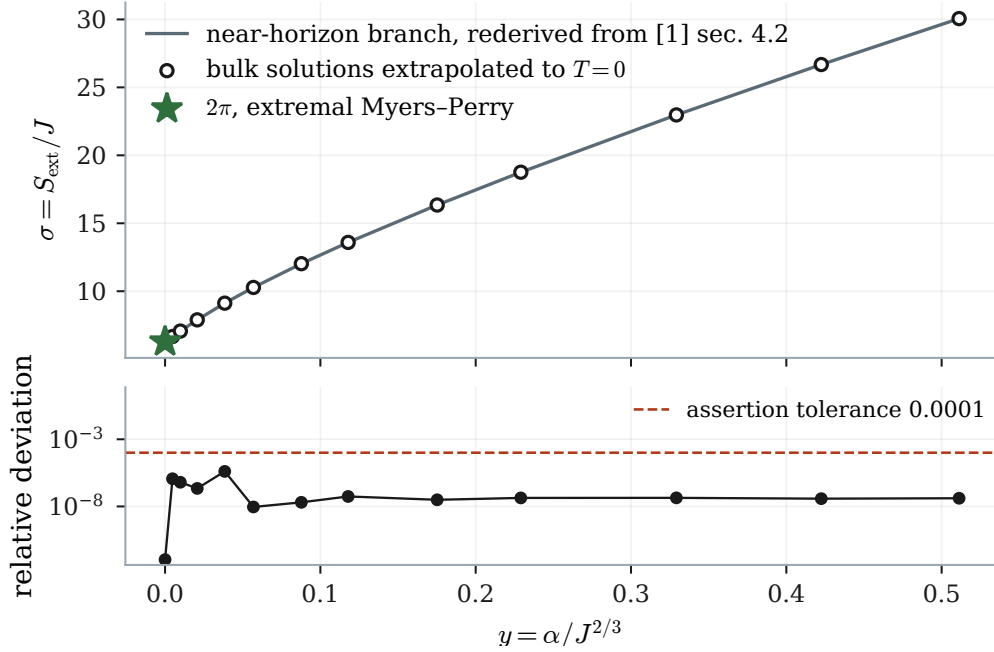


Figure 4. The extremal entropy against the published near-horizon branch. Upper: $\sigma = S_{\text{ext}}/J$ from the near-horizon entropy function, rederived here and checked against eqs. (4.11) and (4.12) of arXiv:1010.0860 in the paper’s own variables, against the zero-temperature extrapolation of the bulk solutions. The star is the extremal Myers–Perry value 2π . Lower: relative difference between the two routes.